TITAN: Analytical FMM/M2L Boundary-Integral Poisson-Boltzmann Solves with Sparse Near-Field ILU(0) Preconditioning

Submission-polished evidence-backed draft

Abstract

Boundary-integral Poisson-Boltzmann solvers can avoid volumetric grids but require accurate acceleration and robust Krylov preconditioning. We present a guarded fourth-order analytical FMM/M2L path in TITAN, built around analytic singular near-field integration, deterministic far-cell refinement, compressed Green-function jets, packed AVX2 direct fallback, and sparse near-field ILU(0). The method is evaluated as an advanced explicit TABI treecode option, not as a default backend. The primary full-protein panel includes Barnase, the RNA-binding domain of E. coli Rho factor (1A63), and Stichodactyla helianthus neurotoxin I (1SH1). All guarded p4 repeats converged with finite energies, sampled direct-operator residuals below 8e-3, and one plan/cache build. Additional public proteins, Ubiquitin (1UBQ), Chymotrypsin inhibitor 2 (2CI2), and Villin headpiece subdomain (1VII), passed the same supplement gate. A dense Barnase convergence profile shows that sparse near-field ILU(0) resolves a high-coupling case where local block-diagonal preconditioning stalls. The production Poisson-Boltzmann default remains the grid backend, and the default TABI treecode path remains degree-one.

Methods

TITAN solves a coupled boundary-integral Poisson-Boltzmann system over triangulated molecular surfaces. The accelerated path follows the FMM motivation of Greengard and Rokhlin [2] and the Cartesian boundary-integral PB context of Chen, Tausch, and Geng [1]. GMRES is used for the nonsymmetric coupled system [3]. The guarded p4 configuration is explicit: Taylor order 4, opening theta 0.6, static tail tolerance 0.002, strict-geometry MAC theta 0.5, analytic-singular near integration, compressed degree-9 Green jets, packed AVX2 direct fallback, and sparse near-field ILU(0).

For panel i, the coupled unknown is u_i = (phi_i, partial_n phi_i)^T. The sparse preconditioner keeps a 2 x 2 block on FMM near-source graph edges, including self edges and at most 64 neighbors. ILU(0) keeps the original graph sparsity and applies block forward/back substitution, following the no-fill incomplete-factorization principle [4]. Duffy-adaptive quadrature [5] remains an explicit near-field option but is not selected automatically.

Protein structures are prepared with PDB2PQR/APBS-family tooling [6], [7] and public structure identifiers from the RCSB Protein Data Bank [8]. Large-system verification uses sampled direct-

operator RMS and sampled direct true residuals because full direct TABI reaction energies are not practical for the largest rows.

Results

The primary guarded p4 panel passed all repeatability gates. Supplementary full-protein systems also passed, broadening the public benchmark context without changing the main claim. The Barnase convergence profile is interpreted narrowly as a dense/high-coupling preconditioner contrast.

Limitations

Grid remains TITAN's production Poisson-Boltzmann default. Guarded p4 is an advanced explicit TABI treecode option and is not selected automatically. Full direct TABI reaction energies are not used as the large-system oracle. Duffy-adaptive near integration is explicit-only. The Barnase preconditioner contrast should not be generalized to all proteins.

Primary Protein Panel

Table 1. Guarded p4 repeatability on the locked full-protein panel.

Protein	PDB/source	Panels	Energy	Audit max	Direct residual max	Energy spread	p4/deg1
Barnase	PyDelPhi benchmark	14942	-21030.83	6.383e-05	6.383e-05	0.000	0.450
RNA-binding domain of E. coli Rho factor	1A63	17250	-247716.08	2.357e-03	2.357e-03	0.000	0.674
Stichodactyla helianthus neurotoxin I	1SH1	6673	-4362.07	4.894e-06	4.894e-06	0.000	0.383

Supplementary Repeatability Context

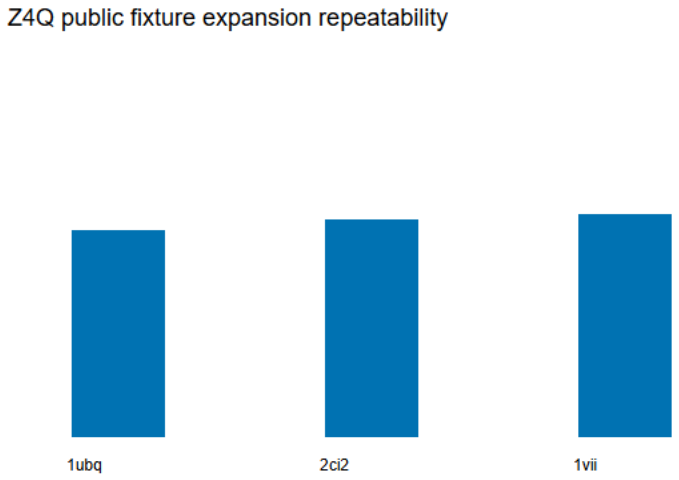


Figure 1. Supplementary public-protein guarded p4 repeatability for 1UBQ, 2CI2, and 1VII.

Preconditioner Convergence Profile

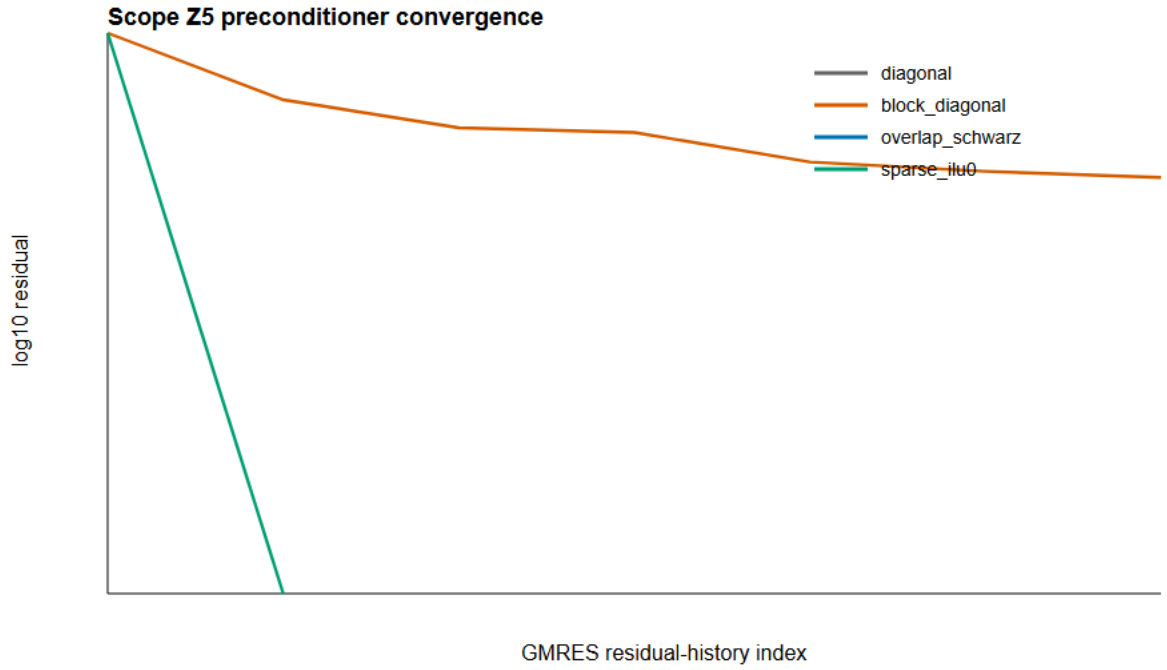


Figure 2. Barnase dense/high-coupling convergence profile comparing local block-diagonal preconditioning with sparse near-field ILU(0).

Supplementary Public-Protein Expansion

Table S1. Full public-protein supplement rows prepared without atom sampling.

Protein	PDB	Atoms	Panels	Energy	Audit max	Direct residual max	Energy spread
Ubiquitin	1UBQ	1231	11242	-4038.46	1.468e-05	1.468e-05	0.000
Villin headpiece subdomain	1VII	596	5032	-2355.70	5.871e-06	5.871e-06	0.000
Chymotrypsin inhibitor 2	2CI2	1076	9889	-4891.15	7.983e-06	7.983e-06	0.000

References

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